1.write a c program to search an element from an array of elements using sequential search technique

```
#include <stdio.h>

int main() {
    int a[100], n, i, key, pos = -1;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter %d elements:\n", n);
    for (i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }

    printf("Enter element to search: ");
    scanf("%d", &key);

    /* Sequential Search */
    for (i = 0; i < n; i++) {
        if (a[i] == key) {
            pos = i;
            break;
        }
    }

    if (pos == -1)
        printf("Element %d not found\n", key);
    else
        printf("Element %d found at position %d\n", key, pos);

    return 0;
}
```

Output:

```
Enter number of elements: 5
Enter 5 elements:
10 25 32 7 18
Enter element to search: 32
Element 32 found at position 2
```

2.write a c program to search an element from an array of n elements using binary search technique

```
#include <stdio.h>

int main() {
```

```
    int a[100], n, key, low, high, mid, i, pos = -1;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter %d elements (sorted):\n", n);
    for (i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }

    printf("Enter element to search: ");
    scanf("%d", &key);

    low = 0;
    high = n - 1;

    /* Binary Search Logic */
    while (low <= high) {
        mid = (low + high) / 2;

        if (a[mid] == key) {
            pos = mid;
            break;
        }
        else if (key < a[mid]) {
            high = mid - 1;
        }
        else {
            low = mid + 1;
        }
    }

    if (pos == -1)
        printf("Element %d not found\n", key);
    else
        printf("Element %d found at position %d\n", key, pos);

    return 0;
}
```

Output:

Enter number of elements: 6
Enter 6 elements (sorted):
5 12 18 27 45 60
Enter element to search: 27
Element 27 found at position 3

3.write a c program to search an element from an array of n elements using interpolation search technique

```
#include <stdio.h>

/* Function for Interpolation Search */
int interpolationSearch(int a[], int n, int key) {
    int low = 0;
    int high = n - 1;
    int pos;

    while (low <= high && key >= a[low] && key <= a[high]) {
        if (low == high) {
            if (a[low] == key)
                return low;
            return -1;
        }

        /* Estimate the position */
        pos = low + ((key - a[low]) * (high - low)) / (a[high] - a[low]);

        if (a[pos] == key)
            return pos;

        if (a[pos] < key)
            low = pos + 1;
        else
            high = pos - 1;
    }

    return -1;
}

int main() {
    int a[100], n, i, key, result;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter %d elements (sorted):\n", n);
    for (i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }

    printf("Enter element to search: ");
    scanf("%d", &key);
```

```
    result = interpolationSearch(a, n, key);

    if (result == -1)
        printf("Element %d not found\n", key);
    else
        printf("Element %d found at position %d\n", key, result);

    return 0;
}
```

Output:

Enter number of elements: 6
Enter 6 elements (sorted):
10 20 30 40 50 60
Enter element to search: 40
Element 40 found at position 3